Generalised Schwartz Spaces Do Not Suffice for Weak-to-Strong Extension

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Source Question and Status

Kruse proves weak-to-strong extension theorems for bounded uniformly continuous functions and for the disc algebra when the target is semi-Montel. He then asks whether those two extension results still hold when the target is merely a generalised Schwartz space which is not semi-Montel [1, Corollaries 4.8–4.9 and the paragraph following them, p. 21].

4.8. **Corollary.** *Let Ω be a metric space, $U \subset \Omega$ a dense subset and E a semi-Montel space. If $f: U \to E$ is a function such that $e' \circ f$ admits an extension $f_{e'} \in \mathcal{C}_{bu}(\Omega)$ for each $e' \in E'$, then there is a unique extension $F \in \mathcal{C}_{bu}(\Omega, E)$ of f. In particular,*

$$\mathcal{C}_{bu}(\Omega, E) = \{ f: \Omega \to E \mid \forall\ e' \in E' : \ e' \circ f \in \mathcal{C}_{bu}(\Omega) \}.$$

4.9. **Corollary.** *Let $\Omega \subset \mathbb{C}$ be open and bounded, $U \subset \Omega$ fix the topology in $\mathcal{A}(\Omega)$ and E a semi-Montel space over $\mathbb{C}$. If $f: U \to E$ is a function such that $e' \circ f$ admits an extension $f_{e'} \in \mathcal{A}(\overline{\Omega})$ for each $e' \in E'$, then there is a unique extension $F \in \mathcal{A}(\overline{\Omega}, E)$ of f. In particular,*

$$\mathcal{A}(\overline{\Omega}, E) = \{ f: \overline{\Omega} \to E \mid \forall\ e' \in E' : \ e' \circ f \in \mathcal{A}(\overline{\Omega}) \}.$$

If E is a generalised Schwartz space which is not a semi-Montel space, we do not know whether the extension results in Corollary 4.8 and Corollary 4.9 hold but we still have a weak-strong principle due to the following observation which is based on [38, Chap. 3, §9, Proposition 2, p. 231] with $\sigma(E, E')$ replaced by $\sigma(E, G)$.

Both answers are negative, already for the same concrete target space. Status: *candidate full counterexample, likely valid*, subject to human review.

Definitions and Proof Intuition

A locally convex space is a *generalised Schwartz space* in the source's terminology when every bounded subset is precompact. A semi-Montel space requires every bounded subset to be relatively compact.

Let

$$E = c_{00} = \{x = (x_k)_{k \geq 1} \in \mathbb{C}^{\mathbb{N}} : x_k = 0 \text{ for all but finitely many } k\}$$

with the topology inherited from the product space $\mathbb{C}^{\mathbb{N}}$. Finite coordinate windows see every neighbourhood and every continuous linear functional. Consequently bounded sets are precompact, but a coordinatewise Cauchy sequence may converge only in the completion $\mathbb{C}^{\mathbb{N}}$. The two counterexamples arrange that all scalar extensions exist while the forced vector value at one missing point has infinitely many nonzero coordinates and hence lies outside E.

For the disc algebra, the extra analytic structure is preserved by using polynomials which successively vanish at a dense list of boundary points.

The Target Space

Lemma 1. *$E = c_{00}$ with its product-subspace topology is a complex generalised Schwartz space, is not semi-Montel, and*

$$E' = \operatorname{span}\{\pi_k : k \geq 1\}, \qquad \pi_k(x) = x_k.$$

Its completion is $\widehat{E} = \mathbb{C}^{\mathbb{N}}$.

Proof. Every zero-neighbourhood in E contains one which restricts only finitely many coordinates. If $B \subset E$ is bounded, its projection onto those finitely many coordinates is bounded in a finite-dimensional space and therefore has a finite net. Lifting the net by zero in all other coordinates shows that B is precompact.

The truncations

$$s_n = (\underbrace{1, \ldots, 1}_{n}, 0, 0, \ldots)$$

form a bounded subset of E and converge in $\mathbb{C}^{\mathbb{N}}$ to $\mathbf{1} = (1, 1, \ldots) \notin E$. If their closure in E were compact, its continuous image in the Hausdorff space $\mathbb{C}^{\mathbb{N}}$ would be closed and would contain $\mathbf{1}$, a contradiction. Thus E is not semi-Montel. Truncation also proves that E is dense in the complete product $\mathbb{C}^{\mathbb{N}}$. Finally, continuity of a linear functional at zero forces it to vanish on every coordinate outside some finite set, which gives the displayed dual. □

Full Negative Result

Theorem 1. *For the generalised Schwartz, non-semi-Montel space E of Lemma 1, both extensions proposed after Corollaries 4.8 and 4.9 of [1] fail.*

1. *There are a compact metric space Ω, a dense subset $U \subset \Omega$, and $f : U \to E$ such that $e' \circ f$ has an extension in $\mathcal{C}_{bu}(\Omega)$ for every $e' \in E'$, but f has no extension in $\mathcal{C}_{bu}(\Omega, E)$.*
2. *There are a set $U \subset \partial\mathbb{D}$ which fixes the topology of the disc algebra $\mathcal{A}(\overline{\mathbb{D}})$ and a map $f : U \to E$ such that $e' \circ f$ has an extension in $\mathcal{A}(\overline{\mathbb{D}})$ for every $e' \in E'$, but f has no extension in $\mathcal{A}(\overline{\mathbb{D}}, E)$.*

Proof. For part 1, take

$$\Omega = \{0\} \cup \{1/n : n \geq 1\}, \qquad U = \{1/n : n \geq 1\}, \qquad f(1/n) = s_n.$$

Every $e' \in E'$ uses only finitely many coordinates, so $e'(s_n)$ is eventually constant. It therefore extends continuously to 0, and the scalar extension is bounded and uniformly continuous because Ω is compact. If $F \in \mathcal{C}_{bu}(\Omega, E)$ extended f, continuity of $\pi_k \circ F$ would give $\pi_k(F(0)) = 1$ for every k. This would force $F(0) = \mathbf{1} \notin E$, a contradiction.

For part 2, let $(\zeta_j)_{j \geq 1}$ enumerate without repetition all roots of unity, and put $U = \{\zeta_j : j \geq 1\}$. This set is dense in $\partial\mathbb{D}$. Hence, for $g \in \mathcal{A}(\overline{\mathbb{D}})$,

$$\sup_{z \in U} |g(z)| = \sup_{z \in \partial\mathbb{D}} |g(z)| = \sup_{z \in \overline{\mathbb{D}}} |g(z)|,$$

so U fixes the disc-algebra topology. Define

$$a_n(z) = \prod_{k=1}^{n} (z - \zeta_k), \qquad f(\zeta_j) = (a_n(\zeta_j))_{n \geq 1}.$$

For $n \geq j$ the product $a_n(\zeta_j)$ contains a zero factor. Thus $f(\zeta_j)$ has finite support and belongs to E. Given $e' = \sum_{n=1}^{N} c_n \pi_n \in E'$, the polynomial $\sum_{n=1}^{N} c_n a_n$ is a disc-algebra extension of $e' \circ f$.

Suppose that $F \in \mathcal{A}(\overline{\mathbb{D}}, E)$ extended f. For each n, $\pi_n \circ F$ and a_n are scalar disc-algebra functions agreeing on the dense boundary set U. They agree on $\partial\mathbb{D}$ by continuity and therefore on $\overline{\mathbb{D}}$ by the maximum principle. In particular,

$$\pi_n(F(0)) = a_n(0) = (-1)^n \prod_{k=1}^{n} \zeta_k \neq 0 \quad \text{for every } n.$$

Hence $F(0)$ has infinite support, contradicting $F(0) \in E$. □

Scope, Novelty Check, and Verification

The theorem refutes the unrestricted replacement of "semi-Montel" by "generalised Schwartz but not semi-Montel" in both source corollaries. It does not classify which individual incomplete generalised Schwartz spaces might retain either extension property. The counterexamples use the full dual E', and the disc-algebra construction preserves the source's topology-fixing hypothesis exactly.

The run's result ledger and attempt index were searched by arXiv id, title, author, and the terms "generalised Schwartz", "semi-Montel", "disc algebra", and "weak-to-strong extension". A bounded arXiv search through 17 August 2026 used the exact open sentence, the source's Corollary 4.8/4.9 labels, and the same core terms. It found the source and related extension papers, but no later paper claiming an answer to this exact question.

The companion script checks finite instances of both coordinate patterns: eventual scalar stabilization in part 1, boundary zeros in part 2, and the nonzero forced center coordinates. These are consistency checks; the proof is exact and does not depend on computation.

Human Review Recommendation

Verify Lemma 1's two topology claims and the fact that the dense boundary set fixes the disc-algebra norm. Then check the coordinatewise uniqueness argument at the missing point in each construction. No external structural theorem beyond the elementary maximum principle is needed.

References

[1] K. Kruse, *Extension of vector-valued functions and sequence space representation*, Bulletin of the Belgian Mathematical Society–Simon Stevin **29** (2022), 307–331; arXiv:1808.05182; doi:10.36045/j.bbms.211009.